

MATH 2790 Fall 2025

Homework 6

Due date: Wednesday, Oct 1

Instructions:

- (a) It is expected to show detailed work for every problem. You are allowed to have a discussion with peers on some challenging questions. But then you should work on your own to finish the assignment.
- (b) All questions in this assignment are from sections 3.2–3.3.

Problems:

1. Question (1) on page 181 of textbook under section 3.2.
2. Question (4) on page 181 of textbook under section 3.2.
3. Question (5) on page 182 of textbook under section 3.2. (Hint: How are PV and the accumulated value connected? What information can we get from their connection?)
4. Question (1) on page 182 of textbook under section 3.3.
5. Question (2) on page 183 of textbook under section 3.3. (Hint: what is the interest rate per payment period? Round the interest rate per payment period to 5 decimal places.)
6. Question (4) on page 183 of textbook under section 3.3. Draw the time diagram first.
7. An annuity has payments of 1000 at the beginning of every quarter for six years. Another annuity has payments of X at the end of each year for five years. At an annual effective interest rate of 6%, the present value of the two annuities are equal. Find X . Round to 4 decimal places.